

111

none (E=0)

$Y_i = E \cdot m_i(A_2, A_1, A_0) \rightarrow E = 0$, so every output is 0

truth table

E	A2	A1	A0	active
1	0	0	0	Y0
1	0	0	1	Y1
1	0	1	0	Y2
1	0	1	1	Y3
1	1	0	0	Y4
1	1	0	1	Y5
1	1	1	0	Y6
1	1	1	1	Y7
0	0	0	0	—

test case log

Capture current state Clear log Copy as text

#	time	E	A2A1A0	active
1	19:48:29	1	111	Y7
2	19:48:36	1	001	Y1
3	19:48:48	0	111	none

Built for CIA III (20 marks). Toggle every input combination at least once, capture each one in the test log, and screenshot the log tables for the "screenshots of working simulation and test cases" section of your report.

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1	0	1	0	Y2
1	0	1	1	Y3
1	1	0	0	Y4
1	1	0	1	Y5
1	1	1	0	Y6
1	1	1	1	Y7
0	0	0	0	—

test case log

Capture current state

Clear log

Copy as text

#	time	E	A2A1A0	active
1	19:48:29	1	111	Y7
2	19:48:36	1	001	Y1
3	19:48:48	0	111	none

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Signal bench

An interactive 4:1 multiplexer and 3:8 decoder. Flip the switches, watch the wires carry each bit, and log test cases straight from the truth table below.

4:1 multiplexer 3:8 decoder

A 3:8 decoder reads a 3-bit address, A2 A1 A0, and drives exactly one of eight outputs, Y0–Y7, high — but only while the enable line E is on. With E off, every output stays low regardless of the address.

schematic

controls

E enable

A2 address bit 2

A1 address bit 1

A0 address bit 0

address (A2 A1 A0)

active output

111

Y7

$Y_i = E \cdot m_i(A_2, A_1, A_0) \rightarrow Y_7 = E \cdot A_2 A_1 A_0 = 1$

truth table

E	A2	A1	A0	active
1	0	0	0	Y0
1	0	0	1	Y1
1	0	1	0	Y2
1	0	1	1	Y3
1	1	0	0	Y4
1	1	0	1	Y5
1	1	1	0	Y6
1	1	1	1	Y7

01 00 11 10

D3 data input 3

S1 select bit 1

S0 select bit 0

select (S1 S0)

01

output Y

1

Y = D0S1'S0' + D1S1'S0 + D2S1S0' + D3S1S0 → currently routing **D1** (S1'S0) to Y

truth table			
S1	S0	selects	Y
0	0	D0	1
0	1	D1	1
1	0	D2	1
1	1	D3	0

test case log				
Capture current state		Clear log	Copy as text	
#	time	D3D2D1D0	S1S0	Y
1	19:47:20	1111	11	1
2	19:47:26	1110	01	1

Built for CIA III (20 marks). Toggle every input combination at least once, capture each one in the test log, and screenshot the log tables for the "screenshots of working simulation and test cases" section of your report.



D2 data input 2 ☒

D3 data input 3 ☐

S1 select bit 1 ☐

S0 select bit 0 ☒

select (S1 S0)
01

output Y
1

$Y = D0S1'S0' + D1S1'S0 + D2S1S0' + D3S1S0 \rightarrow$ currently routing **D1** (S1'S0) to Y

truth table

S1	S0	selects	Y
0	0	D0	1
0	1	D1	1
1	0	D2	1
1	1	D3	0

test case log

Capture current state

Clear log

Copy as text

#	time	D3D2D1D0	S1S0	Y
1	19:47:20	1111	11	1
2	19:47:26	1110	01	1

Built for CIA III (20 marks). Toggle every input combination at least once, capture each one in the test log, and screenshot the log tables for the "screenshots of working simulation and test cases" section of your report.

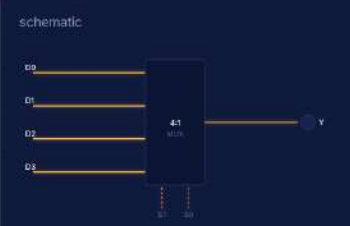
Signal bench

An interactive 4:1 multiplexer and 3:8 decoder. Flip the switches, watch the wires carry each bit, and log test cases straight from the truth table below.

4:1 multiplexer 3:8 decoder

A 4:1 MUX routes one of four data lines, D0-D3, to the single output Y. Two select lines, S1 and S0, choose which one — only one input's wire ever reaches the output at a time.

schematic



controls

D0 data input 0

☐

D1 data input 1

☐

D2 data input 2

☐

D3 data input 3

☐

S1 select bit 1

☐

S0 select bit 0

☐

select (S1 S0)

11

output Y

1

$Y = D0S1'S0' + D1S1'S0 + D2S1S0' + D3S1S0 \rightarrow$ currently routing **D3** (S1S0) to Y

truth table

S1	S0	selects	Y
0	0	D0	1
0	1	D1	1

Signal bench

An interactive 4:1 multiplexer and 3:8 decoder. Flip the switches, watch the wires carry each bit, and log test cases straight from the truth table below.

4:1 multiplexer 3:8 decoder

A 4:1 MUX routes one of four data lines, D0-D3, to the single output Y. Two select lines, S1 and S0, choose which one — only one input's wire ever reaches the output at a time.

schematic



controls

D0 data input 0



D1 data input 1



D2 data input 2



D3 data input 3



S1 select bit 1



S0 select bit 0



select (S1 S0)

00

output Y

1

$Y = D0S1'S0' + D1S1'S0 + D2S1S0' + D3S1S0 \rightarrow$ currently routing **D0** (S1'S0') to Y

truth table

S1	S0	selects	Y
0	0	D0	1
0	1	D1	1

00 01

D2 data input 2

D3 data input 3

S1 select bit 1

S0 select bit 0

select (S1 S0)

00

output Y

0

$$Y = D0S1'S0' + D1S1'S0 + D2S1S0' + D3S1S0 \rightarrow \text{currently routing } \mathbf{D0} (S1'S0') \text{ to } Y$$

truth table

S1	S0	selects	Y
0	0	D0	0
0	1	D1	0
1	0	D2	0
1	1	D3	0

test case log

Capture current state Clear log Copy as text

No test cases captured yet. Toggle the switches above, then capture a state for your report.

Built for CIA III (20 marks). Toggle every input combination at least once, capture each one in the test log, and screenshot the log tables for the "screenshots of working simulation and test cases" section of your report.

Signal bench

To exit full screen, press and hold **Esc**

An interactive 4:1 multiplexer and 3:8 decoder. Flip the switches, watch the wires carry each bit, and log test cases straight from the truth table below.

4:1 multiplexer 3:8 decoder

A 4:1 MUX routes one of four data lines, D0-D3, to the single output Y. Two select lines, S1 and S0, choose which one — only one input's wire ever reaches the output at a time.

schematic



controls

D0 data input 0	☐
D1 data input 1	☐
D2 data input 2	☐
D3 data input 3	☐
S1 select bit 1	☐
S0 select bit 0	☐

select (S1 S0)

00

output Y

0

$Y = D0S1'S0' + D1S1'S0 + D2S1S0' + D3S1S0 \rightarrow$ currently routing **D0** (S1'S0') to Y

truth table

S1	S0	selects	Y
0	0	D0	0